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Low-cost protocol for the construction of prime editing vector for zinc biofortification in rice

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Zinc (Zn) deficiency is a prevalent nutritional problem in regions that mainly rely on plant-based diets. Biofortification has emerged as a viable strategy for addressing this concern in vulnerable populations by improving zinc content in staple foods such as rice. Genome editing technology is crucial in advancing biofortification efforts through inducing precise allelic modification of a target trait. Among these platforms, prime editing is increasingly becoming popular due to its capacity to perform insertions, deletions, and a wide range of base-to-base conversions. Since many economically important traits in rice are controlled by genes wherein allelic differences arise from base substitutions, prime editing presents as a powerful tool for accelerating trait improvement. In *Arabidopsis*, bZIP19 transcription factor, belonging to the F group of BASIC LEUCINE-ZIPPER (F-bZIP) proteins, is one of the central transcriptional regulators activated in Zn-deficiency response. It was demonstrated that the amino acid substitution of Cysteine (C) with Alanine (A) in a Zn sensor motif (ZSM) within bZIP19 protein enhances Zn accumulation in the plant and seed. To facilitate zinc biofortification in rice, we describe a low-cost strategy for the construction of a prime editing vector to induce a nonsynonymous mutation in rice transcription factor bZIP19. For the construction of a prime editing vector, some genetic elements, such as pegRNA spacer, scaffold, RT template, PBS, RNA Pol III promoter, nicking gRNA spacer, need to be added to the backbone vector. In contrast to acquiring long oligonucleotides or gene synthesis commercially, this protocol used a combination of dual asymmetrical PCR (DA-PCR) and overlap extension PCR (OE-PCR) to synthesize long oligonucleotides at a lower cost and free of error, followed by two successive assembly steps facilitated by the use of Type IIS restriction enzymes BsaI and AarI. The assembled vector aims to be used for generating stable rice transformants with increased Zn content through *Agrobacterium*-mediated transformation.